Lecture 8: Planar Graphs, the Mesh Matrix and Mesh Currents

EE201 – Circuit Theory I

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current, meshes voltage nodes

From nodes to meshes

Earlier, from the incidence matrix A:

$$A\mathbf{i} = \mathbf{0} \quad \text{(KCL)}, \qquad \mathbf{v} = A^T\mathbf{e} \quad \text{(KVL)}$$

With the $n-1$ node voltages as unknowns, KVL holds automatically; only KCL is left.

From nodes to meshes

Earlier, from the incidence matrix A:

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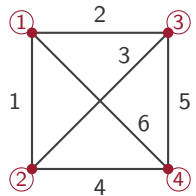
Today: the counterpart, from the meshes

A mesh matrix M with KVL as $M\mathbf{v} = \mathbf{0}$, and mesh currents for which KCL holds automatically.
It works for graphs that can be drawn on a plane without crossings.

Planar graphs

Planar graph

Can be drawn on a plane so that no two branches intersect at a point that is not a node.



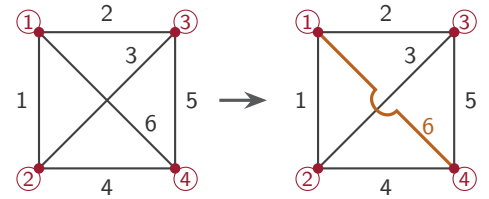
square: 3 and 6 cross

On paper, branches 3 and 6 seem to cross. Do they meet?

Planar graphs

Planar graph

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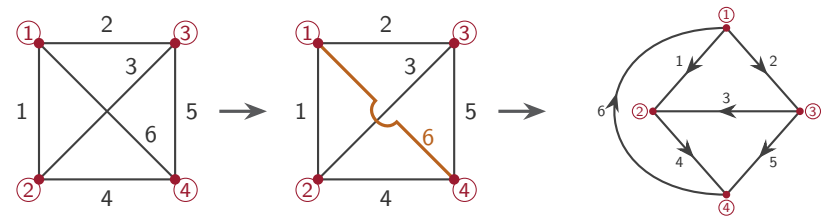
square: 3 and 6 cross **with a hop: 6 over 3**

No: two separate wires, the crossing point is not a node.
In hardware: a jumper wire, or a track on a second layer of the board.

Planar graphs

Planar graph

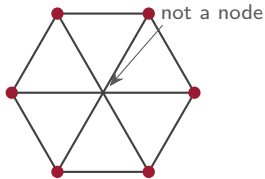
Can be drawn on a plane so that no two branches intersect at a point that is not a node.



Redrawn with no crossing (this is $\mathcal{N}$), so the graph is planar.

A drawing with crossings does not decide it — the question is whether *some* drawing has none.

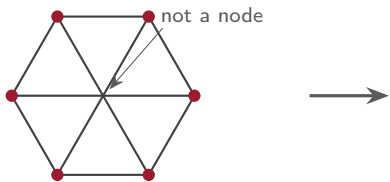
A nonplanar graph



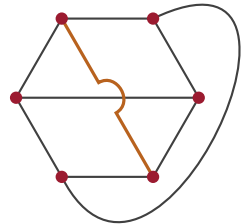
hexagon with its three long diagonals

Three diagonals cross at the centre. Can the graph be redrawn without crossings?

A nonplanar graph



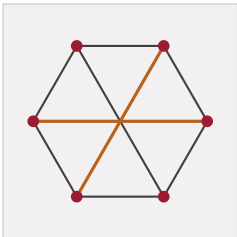
hexagon with its three long diagonals



one diagonal outside: still one hop

However it is redrawn, at least one hop remains: nonplanar.

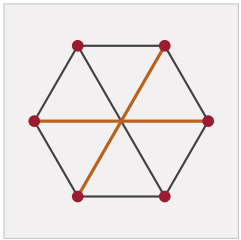
A nonplanar graph in 3D



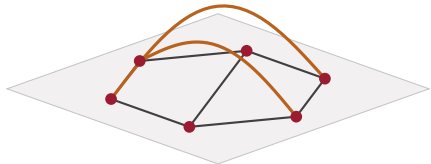
from above

From above it looks like crossings.

A nonplanar graph in 3D



from above



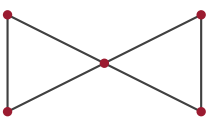
an oblique view

Two diagonals are lifted: no two branches meet in space.
A hop is a step into the third dimension — on a plane, some step is always needed.

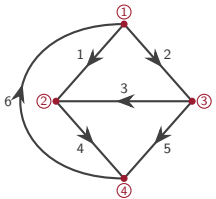
Nonseparable graphs

Nonseparable graph

A connected graph that stays connected after removing any one node (and the branches attached to it).



separable

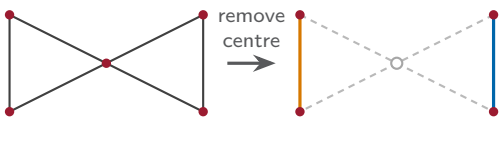


nonseparable ($\mathcal{N}$)

Nonseparable graphs

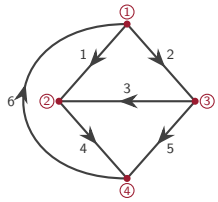
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separable

two pieces

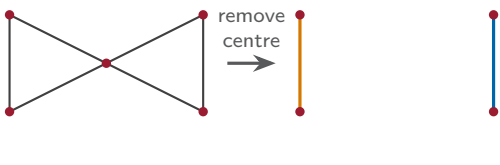


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Nonseparable graphs

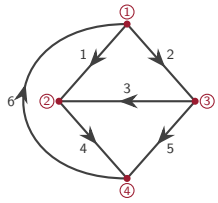
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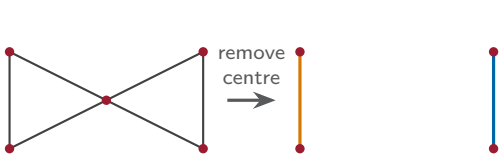


nonseparable ($\mathcal{N}$)

Nonseparable graphs

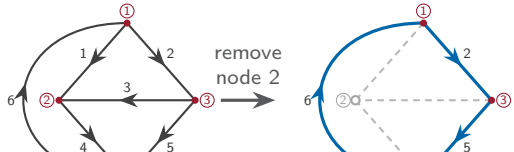
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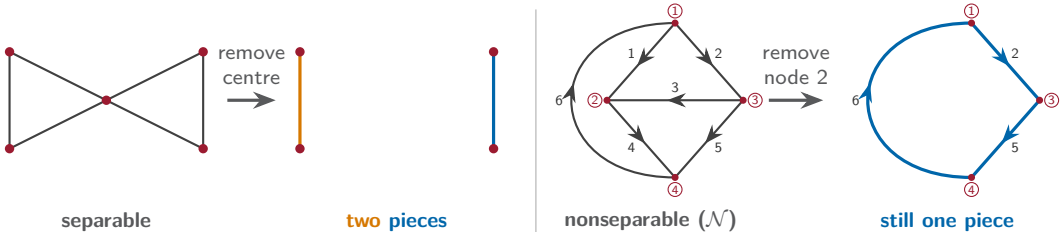
nonseparable ($\mathcal{N}$)

still one piece

Nonseparable graphs

Nonseparable graph

A connected graph that stays connected after removing any one node (and the branches attached to it).

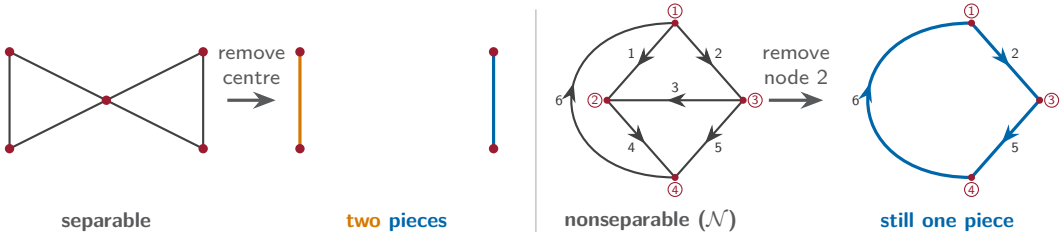


Any node of $\mathcal{N}$ can be removed: the rest stays connected.

Nonseparable graphs

Nonseparable graph

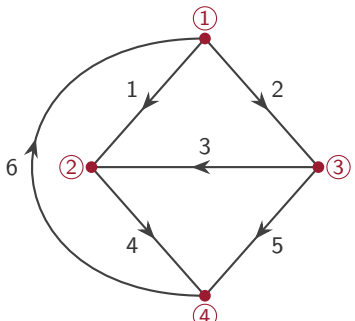
A connected graph that stays connected after removing any one node (and the branches attached to it).



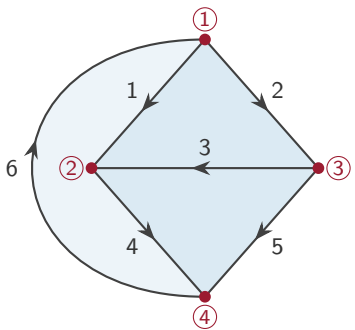
From now on: the graph is planar and nonseparable, drawn without crossings.

Meshes

A planar drawing divides the plane into regions.



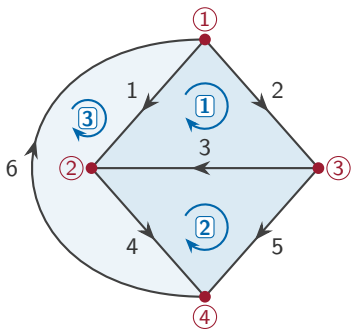
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Inner mesh: the loop around a region inside the graph, with no branch inside it. (The outside region is not counted.)

Meshes



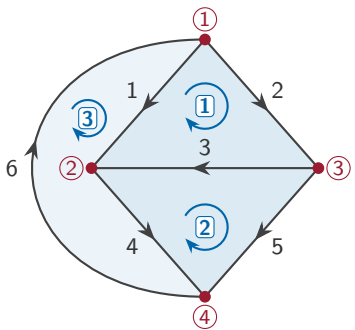
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1: branches 1, 2, 3 2: 3, 4, 5 3: 1, 4, 6

Direction: clockwise, by convention.

Meshes



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Number of inner meshes:
$\ell = b - n + 1 = 6 - 4 + 1 = 3$
— the number of independent KVL equations found earlier.

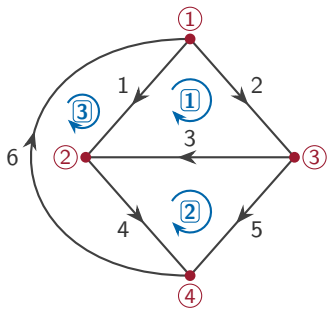
The mesh matrix

Mesh matrix M $(\ell \times b)$

$$m_{jk} = \begin{cases} +1 & \text{branch } k \text{ in mesh } j, \text{ directions coincide,} \\ -1 & \text{branch } k \text{ in mesh } j, \text{ directions opposite,} \\ 0 & \text{otherwise.} \end{cases}$$

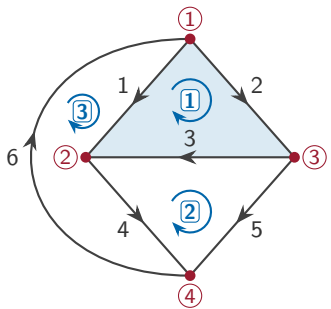
A row is a mesh: walk around it clockwise.
Each branch you meet: $+1$ if its arrow points the way you walk, -1 if it points against you.

Example: the mesh matrix of $\mathcal{N}$



$$M = \begin{array}{c} \\ \boxed{1} \\ \boxed{2} \\ \boxed{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccccc} & & & & & \\ & & & & & \\ & & & & & \end{array}\right] \end{array}$$

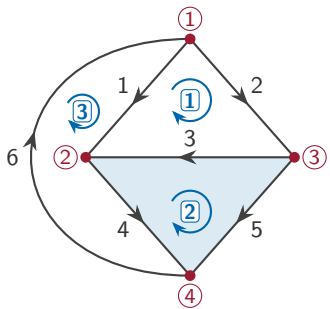
Example: the mesh matrix of $\mathcal{N}$



1: 1 → 3 → 2 → 1. Branches 2, 3 along: $+1$; branch 1 against: -1.

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccccc} -1 & 1 & 1 & 0 & 0 & 0 \\ & & & & & \\ & & & & & \end{array}\right] \end{array}$$

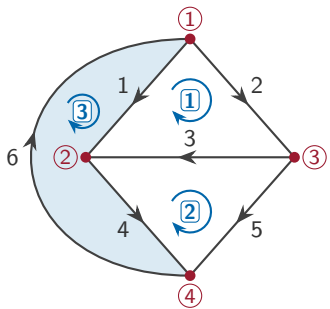
Example: the mesh matrix of $\mathcal{N}$



2: ② → ③ → ④ → ②. Branch 5 along: $+1$; branches 3, 4 against: -1.

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ & & & & & \end{bmatrix} \end{array}$$

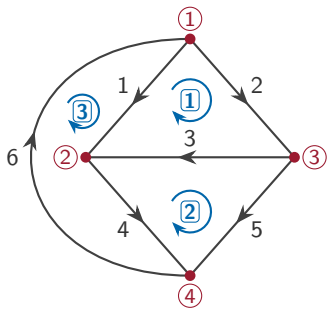
Example: the mesh matrix of $\mathcal{N}$



3: ④ → ① → ② → ④. Branches 6, 1, 4 all along: $+1$.

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

Example: the mesh matrix of $\mathcal{N}$



Three meshes, six branches: M is 3×6.

$$M = \begin{array}{c} \\ \boxed{1} \\ \boxed{2} \\ \boxed{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

KVL with the mesh matrix

$$M\mathbf{v} = \mathbf{0}$$

KVL around every mesh, $\mathbf{v} = [\, v_1 \;\; \cdots \;\; v_b \,]^T$

KVL with the mesh matrix

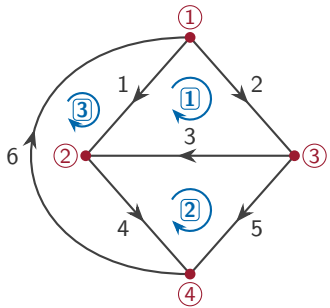
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KVL around every mesh, $\mathbf{v} = [\, v_1 \;\; \cdots \;\; v_b \,]^T$

For $\mathcal{N}$: $\quad -v_1 + v_2 + v_3 = 0, \qquad -v_3 - v_4 + v_5 = 0, \qquad v_1 + v_4 + v_6 = 0$

Walking along an arrow goes from $+$ to $-$: that voltage counts with $+$.

Mesh currents

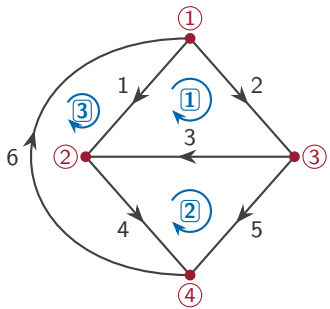


Mesh current

A current i_{Mk} that circulates around mesh k, clockwise.

$\mathbf{i}_M = [\, i_{M1} \;\; \cdots \;\; i_{M\ell} \,]^T$

Mesh currents



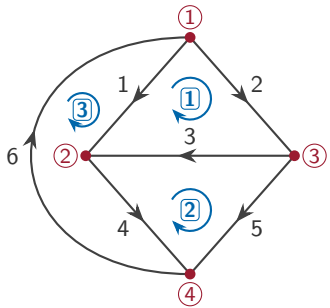
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A branch carries the mesh currents of its meshes: $+$ if the mesh goes along its arrow, $-$ if against.

Mesh currents



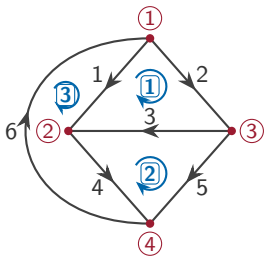
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A branch carries the mesh currents of its meshes: $+$ if the mesh goes along its arrow, $-$ if against.

Branch 3, in 1 (along) and 2 (against):
$i_3 = i_{M1} - i_{M2}$
Branch 2, only in 1: $i_2 = i_{M1}$

Example: branch currents of $\mathcal{N}$ from mesh currents

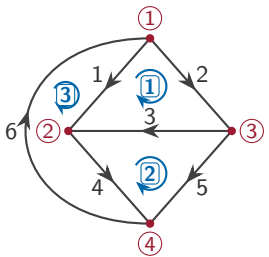


$i_1 =$ $i_4 =$

$i_2 =$ $i_5 =$

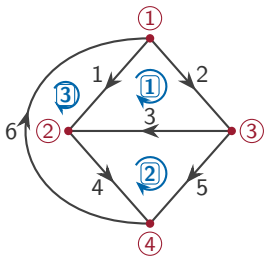
$i_3 =$ $i_6 =$

Example: branch currents of $\mathcal{N}$ from mesh currents



$$i_1 = -i_{M1} + i_{M3} \qquad i_4 = -i_{M2} + i_{M3}$$
$$i_2 = i_{M1} \qquad i_5 = i_{M2}$$
$$i_3 = i_{M1} - i_{M2} \qquad i_6 = i_{M3}$$

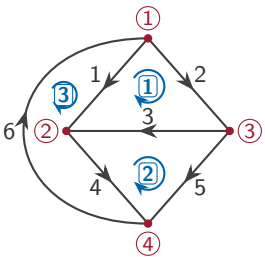
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$$i_2 = i_{M1} \qquad i_5 = i_{M2}$$
$$i_3 = i_{M1} - i_{M2} \qquad i_6 = i_{M3}$$

$$\begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{M^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}$$

Example: branch currents of $\mathcal{N}$ from mesh currents

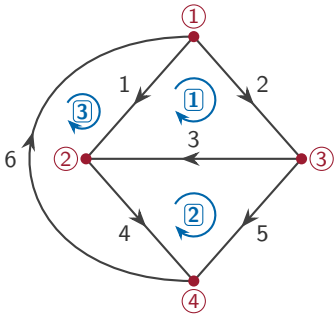


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$$i_3 = i_{M1} - i_{M2} \qquad i_6 = i_{M3}$$

$$\begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{M^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}$$

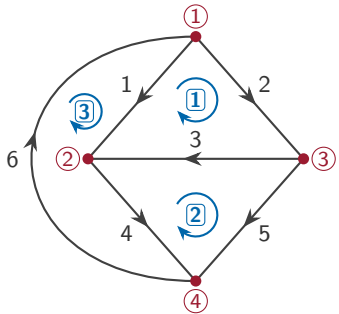
This matrix is M^T: its row 1, $[-1 \;\; 0 \;\; 1]$, is column 1 of M — and so on.

Mesh currents: remarks



$$\mathbf{i} = M^T \mathbf{i}_M$$

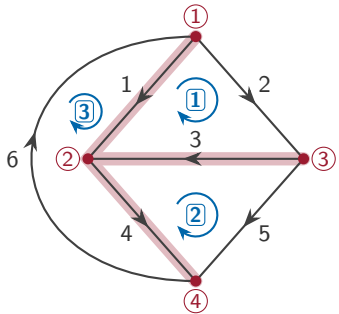
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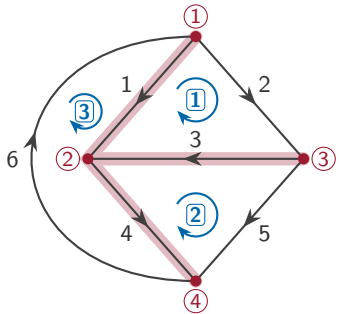


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$$-i_1 - i_3 + i_4 = (i_{M1} - i_{M3}) - (i_{M1} - i_{M2}) + (-i_{M2} + i_{M3}) = 0$$

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- $A\mathbf{i} = AM^T\mathbf{i}_M = \mathbf{0}$ for every $\mathbf{i}_M \Rightarrow AM^T = \mathbf{0}$.

Rows of $A \perp$ rows of M

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$$M = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccccc} -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{array}\right] \end{array} \begin{array}{c} \\ \mathbf{m}_1^T \\ \mathbf{m}_2^T \\ \mathbf{m}_3^T \end{array}$$

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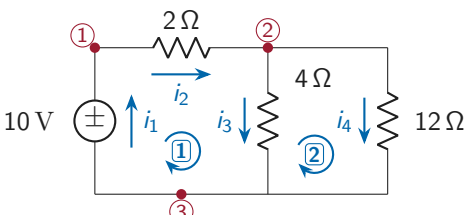
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Node ② and mesh ① share branches 1 and 3: one term $+1$, one -1.

Example: mesh currents in a familiar circuit

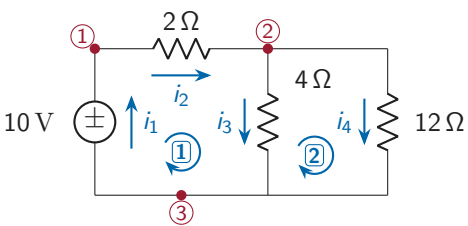
Solved earlier with node voltages: $i_1 = i_2 = 2\,\mathrm{A}$, $i_3 = 1.5\,\mathrm{A}$, $i_4 = 0.5\,\mathrm{A}$, $\mathbf{v} = [-10 \;\; 4 \;\; 6 \;\; 6]^T\,\mathrm{V}$.
Write M and check both results with mesh currents.



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1: up the source, right through 2 Ω, down the 4 Ω — all along.

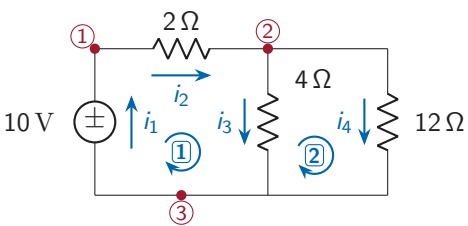
2: up the 4 Ω (against), down the 12 Ω (along).

$$M = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

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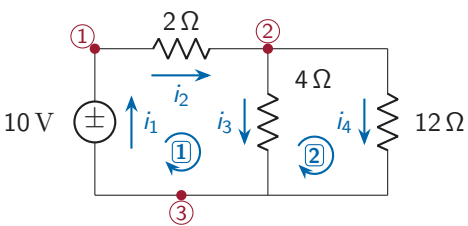
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$i_{M1} = 2\,\text{A}$, $i_{M2} = 0.5\,\text{A} \Rightarrow$ the same branch currents.

KVL: $M\mathbf{v} = [-10 + 4 + 6, \;\; -6 + 6]^T = \mathbf{0}$

Node voltages and mesh currents

	node voltages	**mesh currents**
works for	any connected graph	planar, nonseparable
number of unknowns	$n-1$	$\ell = b-n+1$
matrix	$A \;\; ((n-1)\times b)$	$M \;\; (\ell \times b)$
holds automatically	KVL: $\mathbf{v} = A^T\mathbf{e}$	KCL: $\mathbf{i} = M^T\mathbf{i}_M$
left to write	KCL: $A\mathbf{i} = \mathbf{0}$	KVL: $M\mathbf{v} = \mathbf{0}$

And the two matrices fit together: $AM^T = \mathbf{0}$.

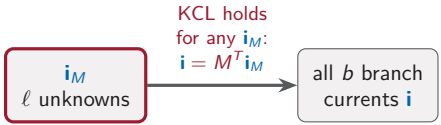
Looking ahead: why mesh currents?

Same idea, KCL and KVL swapped: b element, $n-1$ KCL, ℓ KVL equations.

$\mathbf{i}_M$
ℓ unknowns

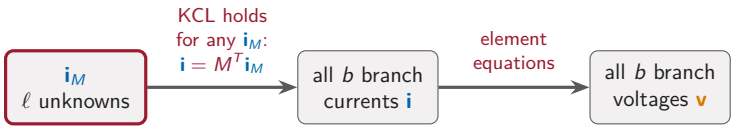
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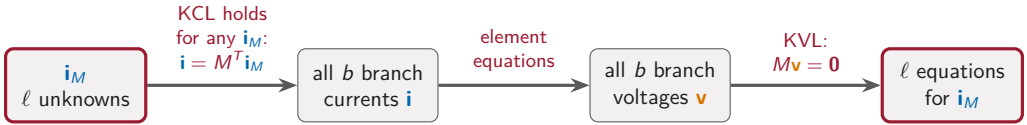
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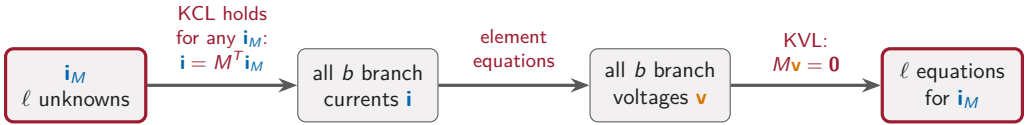
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$2b$ unknowns $\rightarrow \ell = b - n + 1 < b$. Fewer: $n-1$ or ℓ? It depends on the circuit.

Mesh analysis — coming soon.

Summary

- **Planar**: can be drawn with no crossings. **Nonseparable**: removing any one node leaves it connected.
- **Inner meshes**, clockwise: $\ell = b - n + 1$ of them.
- **Mesh matrix** M ($\ell \times b$): $+1$ along, -1 against. KVL: $M\mathbf{v} = \mathbf{0}$.
- **Mesh currents**: $\mathbf{i} = M^T \mathbf{i}_M$. No constraint from KCL, so $AM^T = \mathbf{0}$.
- **Why**: with $\mathbf{i}_M$ as unknowns KCL holds automatically; ℓ unknowns are enough — the idea of mesh analysis.

Next lecture

Nonplanar circuits

Trees, cutsets and fundamental cutsets — tools that work for any connected graph.

Questions?